29	**D**	When each electron is accelerated through the electric field, increase in K.E = decrease in E.P.E. $\text{final K.E. of electron} - 0 = \lvert -e\Delta V \rvert$ The X-ray photon with the shortest wavelength is produced when an electron loses all its K.E. (maximum loss in K.E. possible) to form the highest energy X-ray photon. $\text{final K.E. of electron} - 0 = \dfrac{hc}{\lambda_{\text{min}}}$ $\lambda_{\text{min}} = \dfrac{hc}{\lvert -e\Delta V \rvert} = \dfrac{(6.63\times10^{-34})(3.00\times10^{8})}{(1.60\times10^{-19})(30\times10^{3})} = 4.1438\times10^{-11} = 4.1\times10^{-11}\ \text{m}$
30	**C**	initial count rate of source alone, $C_0 = 100 - 20 = 80\ \text{s}^{-1}$